

GENO-MAP: Correspondence-Free Diagnostics for Sweet Potato Diversity Maps

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Correspondence-free validation of disjoint DArT panels via geometry diagnostics and robustness curves; PCA is the stable analytic baseline in $n \ll p$.

1. Why Correspondence-Free?

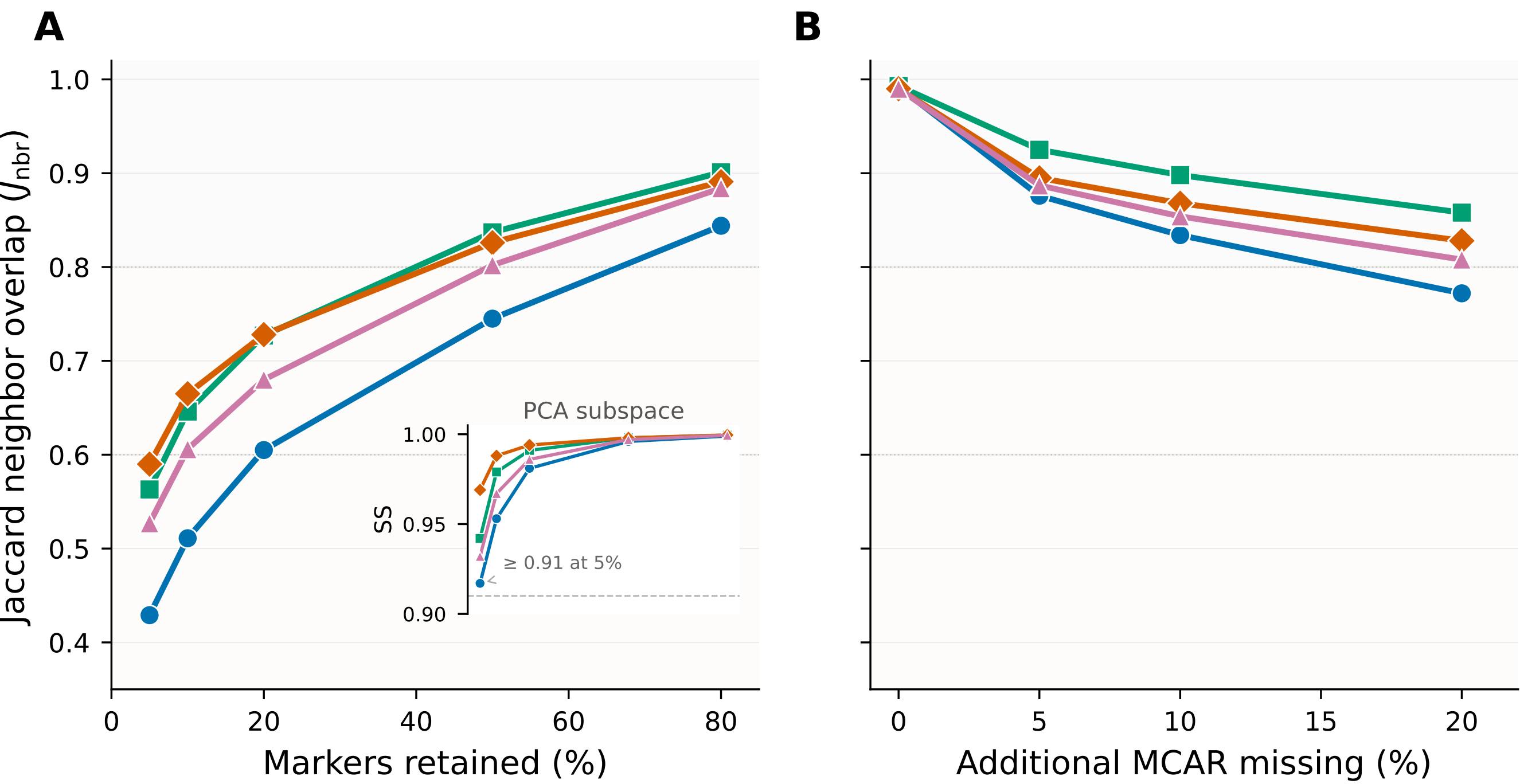
Sweet potato DArT panels use **disjoint ID namespaces** (CIP accessions vs. plate/well coordinates). Without an external manifest, cross-panel alignment is infeasible. We validate each panel *independently* via geometry diagnostics and robustness curves.

Panel	n	p	n/p
G-SNP	5970	20 069	0.30
G-Silico	5970	57 715	0.10
LD-SNP	630	62 732	0.01
LD-Silico	635	38 272	0.02

2. Method (fixed graph protocol)

- PCA-30D** — deterministic analytic space; retains $> 85\%$ variance across all panels.
- kNN graph** — $k = 15$, cosine distance on PCA scores; reciprocity > 0.64 (all panels).
- UMAP-2D** — stochastic layout for visualisation only (not used in metrics).
- Validation** — geometry diagnostics + robustness perturbations (marker dropout 5–80%, MCAR injection +0–20%).

3. Robustness Under Perturbation



Global geometry is invariant. Under marker subsampling (5–80%) and +0–20% MCAR injection, the PCA subspace similarity remains $SS \geq 0.91$ (even at 5% marker retention) and $SS > 0.998$ under missing injection. Principal variance structure is genome-wide, not marker-specific.

Local topology degrades smoothly. kNN neighborhood overlap (J_{nbr}) decreases monotonically from 0.90 to 0.43 under subsampling and to 0.77–0.86 at +20% MCAR.

No phase transitions. Degradation is continuous across all panels, enabling curator-defined quality thresholds without structural collapse.

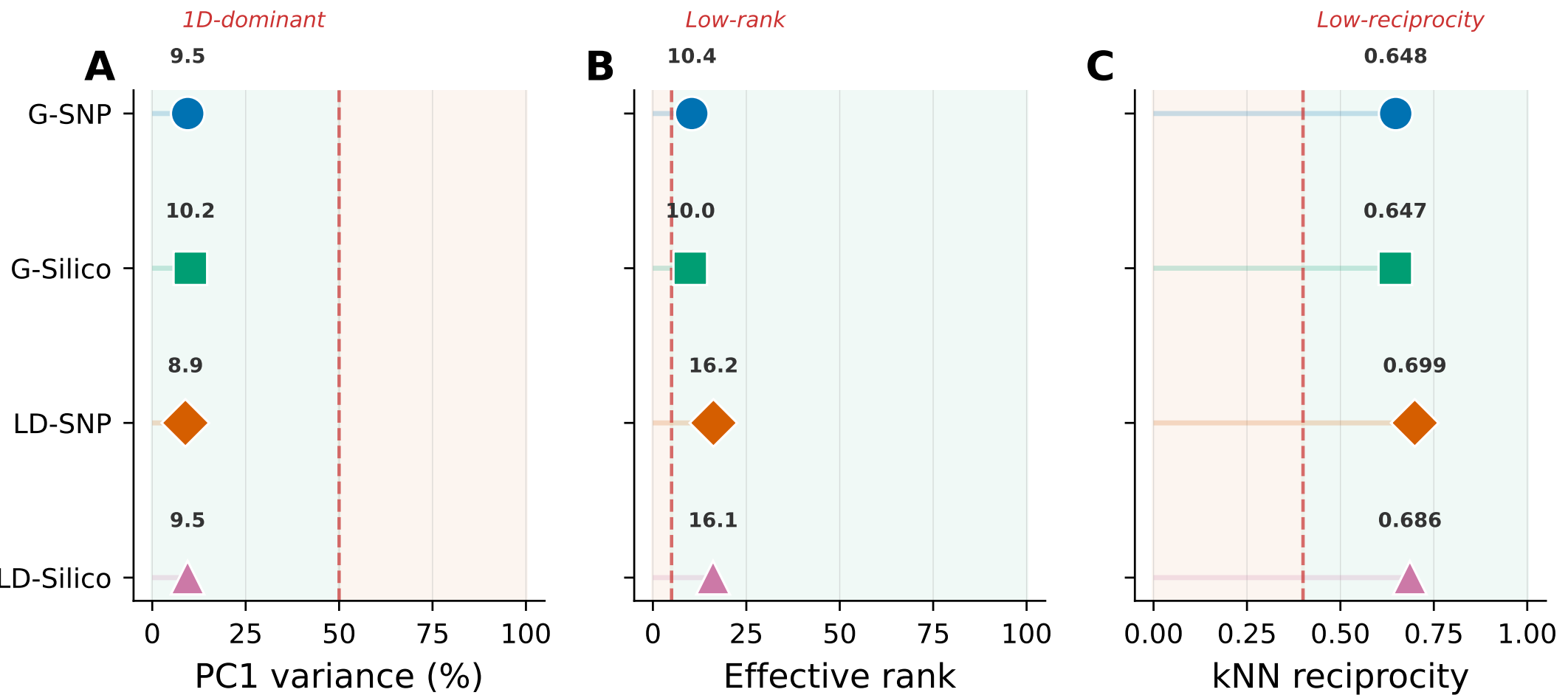
Stability Regime

Let $X \in \mathbb{R}^{n \times p}$ with $n \ll p$. For random marker subsets X_s with $|s| \geq 0.05p$,

$$\mathcal{S}_{30}(X) \approx \mathcal{S}_{30}(X_s) \quad (SS \geq 0.91)$$

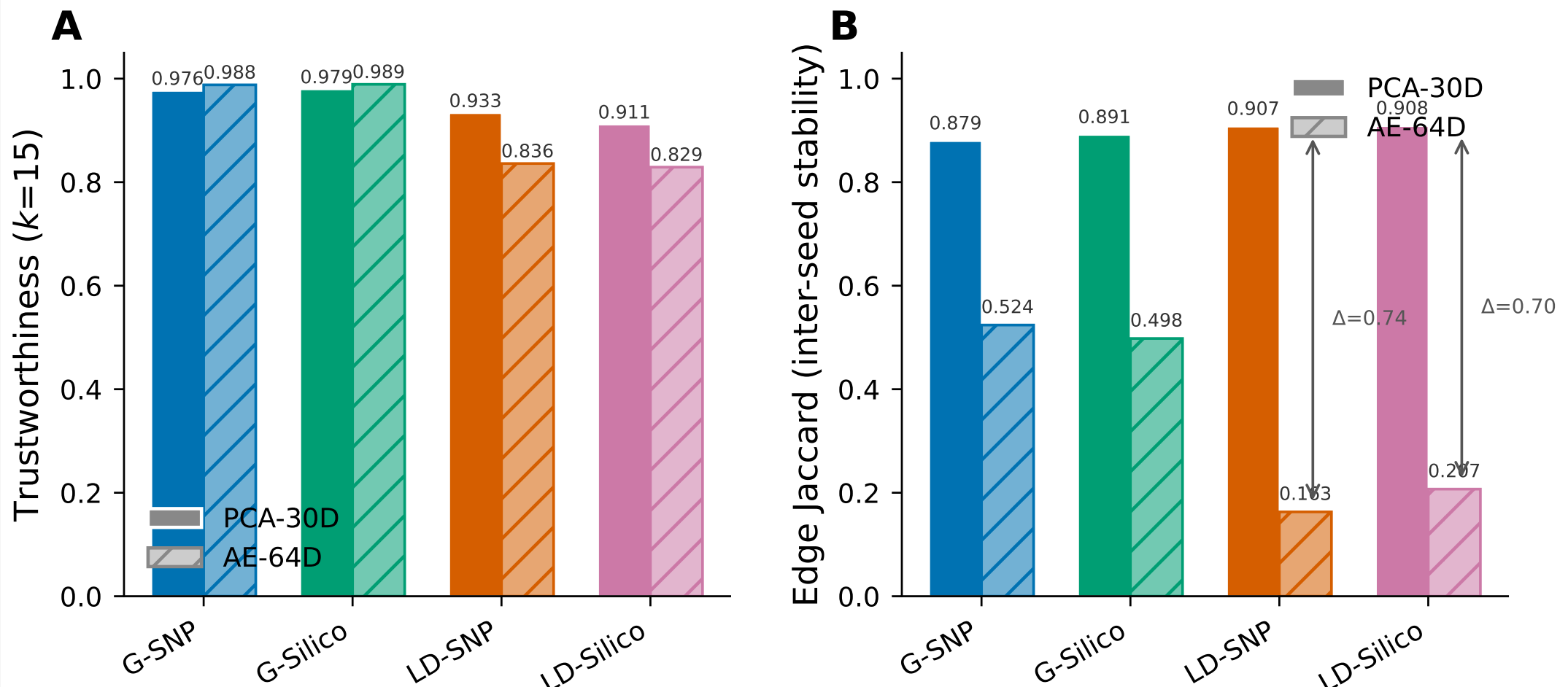
Global PCA geometry is stable under extreme feature dropout, while local neighborhoods degrade continuously without collapse.

4. Panel Geometry & QA



QA thresholds (dashed): $PC1 > 50\%$, $eff. rank < 5$, $reciprocity < 0.40$. All panels show **no 1D collapse** and **strong reciprocity**.

5. PCA vs. Autoencoder

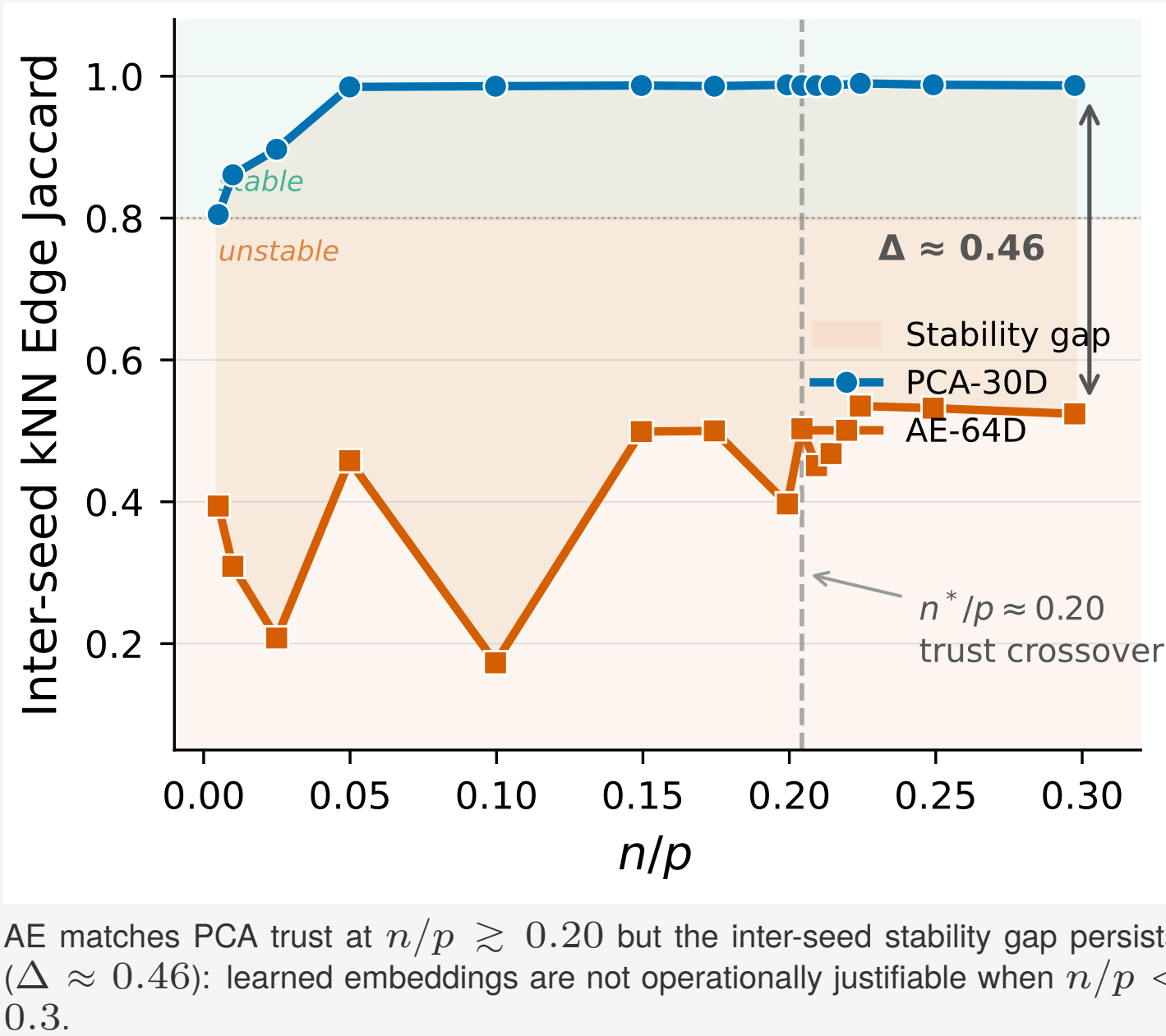


AE-64D yields $\leq +1$ pp trust on Global panels but loses **8–10 pp** on LowDensity. Edge-Jaccard stability: PCA **0.88–0.91**; AE **0.16–0.52**.

Synthesis

- Subspace invariance:** PCA-30D geometry remains stable ($SS \geq 0.91$) under 95% marker dropout.
- Topological continuity:** kNN neighborhoods degrade monotonically without phase transitions.
- Correspondence-free QA:** Geometry diagnostics + robustness curves provide intra-panel validation without shared identifiers.
- Representation constraint:** In $n \ll p$ regimes, stability dominates marginal trust gains; PCA remains operationally preferable.

6. Stability Regime



AE matches PCA trust at $n/p \gtrsim 0.20$ but the inter-seed stability gap persists ($\Delta \approx 0.46$): learned embeddings are not operationally justifiable when $n/p < 0.3$.

7. Failure Mode

In ultra-wide regimes ($n/p < 0.2$), learned embeddings exhibit high inter-seed variance, producing **non-reproducible kNN graphs** and unstable downstream community structure.

Indicator	$n/p < 0.2$
AE seed variance	high
Graph topology	unstable
Clustering reprod.	non-repr.

Robust geometry enables reliable exploration.



Code and reproducible artifacts